

Cadence Installation Guide

Product Version 5.0

January 2024

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Printed in the United States of America.

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About This Guide

The *Cadence Installation Guide* provides the basic information and procedures required to install Cadence® products. Though [Chapter 8, "Overview of Software Configuration"](#) provides a brief summary of software configuration, refer to your product documentation for detailed configuration information.

To use this guide effectively, you need to determine your level of experience with Cadence product installation and system administration. If you have installed Cadence products earlier and have some experience with system administration, you may use the *Cadence Installation Quick Reference* that is provided as a DVD insert. Otherwise, you should read the entire guide.

Additional Information

Cadence Installation Quick Reference

The *Cadence Installation Quick Reference* is a printed document included in the DVD case. It briefly outlines how to install Cadence products. It contains no details on licensing, configuration, or testing.

InstallScape User Guide

The InstallScape User Guide containing information on how to configure and use InstallScape is available at downloads.cadence.com → DOWNLOAD ASSISTANTS->InstallScape and also at the top level of your product Media/DVD. It is strongly recommended that you go through this guide and familiarize yourself with InstallScape before starting installation of Cadence software.

Release-Specific Information

A `README` file or `README.S` directory containing release-specific or product-specific information might exist for your product. If this file exists, you will find it at the top level of your product Media/DVD.

Customer Support Contact Information

For customer support contact information, go to

www.cadence.com/support

Online Support Portal

For online technical support, go to the Cadence Online Support web site:

<https://ask.cadence.com>

The Cadence Online Support web site provides important product-related documents. This service is available to all Cadence customers who have a software support services agreement.

Click *Installation & Licensing*. The page that appears contains useful installation and licensing related articles.

Customer Training

For information about customer training, go to

www.cadence.com/education

General Information

For general information on Cadence Design Systems or Cadence products, go to

www.cadence.com

About Cadence Installation Software

Cadence uses the following installation programs for various delivery and platform options:

- **InstallShield** installs on Microsoft Windows platform.
- **InstallScape** installs on UNIX/Linux platform.

Although these two installation programs are different, they follow the same basic format:

1. Mount or locate your installation media.
2. Run the installation program.
3. Follow the on-screen instructions to install the Cadence software.
4. Perform any post-installation tasks described in your product documentation.

Note: In the case of an acquisition, the software installer may not be in InstallScape or InstallShield. Follow the release-specific release note for installation guide.

Software Delivery

Cadence uses the following software delivery methodology:

- Individual files are grouped into larger collections called *kits*. These kits contain additional information used for tasks such as configuration control and file integrity checking.
- Kits are combined in various ways to create products and allow products to share functionality.
- Each kit may contain special configuration code supplied by the product team.

Cadence has developed utilities to manage these kits and the relationships of these kits between the different products.

The Cadence installation software applies updates to the software using the Update/Hotfix process (for information regarding Update and Hotfix, see [step 2](#) on page 11). By managing the inter-product and intra-product dependencies, the installation software updates all products in a

coordinated fashion.

Installing Software Online

If you are authorized for electronic software distribution, you can download a software Media image from Cadence download.cadence.com. Alternatively, you can use InstallScape application to connect to Cadence downloads server for download and/or install of the software.

If you are installing Cadence software using a downloaded Media Image method, each Media downloaded is a tar file (Windows software media is a zip), which must be unpacked with the `tar` (or equivalent compatible) command. The tar file is not compressed. After you have unpacked the files, use InstallScape or InstallShield to install the Cadence software on UNIX/Linux or Windows systems, respectively. See [step 7](#) on page 16 for more information.

If you are installing Cadence software using InstallScape to connect to Cadence server, refer to the InstallScape documentation located at download.cadence.com → DOWNLOAD ASSISTANTS->InstallScape.

Important Reminders

- Always use Cadence installation software to install and update your Cadence application software. If you modify or install application software without using the Cadence installation software, the data files used to determine the status of each product can be made obsolete.
- Always use the installation software shipped with the application software you are installing. Older versions might not work with the latest control files.
- Always install products from different releases into separate directories. To ensure that different releases do not corrupt the installation data files, the Cadence installation software does not let you install products from different releases into the same directory.

Before You Begin the Installation

Before installing any Cadence products,

1. Check the Cadence software emails and its contents.

The emails from `edaontapsystem@cadence.com` contains

- An e-mail message of the Cadence License File
- An e-mail message of the Cadence Control File

These two emails are sent simultaneously as your order is fulfilled.

Note: Not all orders contain an Installation Control file. This is because Cadence has been using a new order entry/tracking system over the past few years which no longer generates an installation control file. Control file is only intended for software built as InstallScape and InstallShield.

The subject of the e-mail containing the Installation Control file has the following format:

```
#INSTALLSCAPE PRODUCT "Product_ID" "Production Description"
```

```
#INSTALLSCAPE SOURCE "sourceURL" "Description" "Version"
```

For example, a partial entry of InstallScape control file for IC231 is shown below

```
#INSTALLSCAPE PRODUCT "P95100" "P95100: Virtuoso(R) Schematic Editor L"
```

```
#INSTALLSCAPE SOURCE "is/IC231/lnx86/Base" "Base IC231 LINUX RHEL 7.4;RHEL  
7;RHEL 8;SLES 12;SLES 15 IC23.1.43 Release" "Version Base IC231 LINUX RHEL 7.4;RHEL  
7;RHEL 8;SLES 12;SLES 15 IC23.1.43 Release"
```

Note: By default the software installation will install all Products packaged from the Media/DVD. When you install the entire software, you will have the utilities/tools to run the

purchased software aka product(s).

2. Determine the type of release received by looking on the DVD label below the words "electronic design software."

There are three different types of releases. Different product families have different release schedules, and your product may use a combination of these release types:

- Base

A complete software release.

- Update

A periodic update of a base software release.

- Hotfix

A collection of critical fixes, released intermittently between regularly scheduled releases.

Note: some hotfix may be built as a complete software release.

1. Go to <https://ask.cadence.com>

- a. If you already have a Cadence Online Support account, enter your login name and password, and click *Log In* to log into your account.

If you do not have a Cadence Online Support account, click *Register Now*, and follow the instructions to create a new account.

- b. Select *Installation & Licensing*

The page that appears shows important information about configuration, system patches, and other requirements that relate to all Cadence products.

- c. Select *Software - Computing Platforms*, and perform the following steps:

- Click *Recommended platform patches for systems running Cadence products* to view a list of recommended OS patches for running Cadence software.
- Click *Supported Platform Matrix for Cadence Applications* to view the supported hardware and operating system matrix to Cadence applications.

Check your system hardware and software configuration to make sure it is properly set up before you attempt to load the Cadence software onto your system. The Cadence family of products runs on a variety of UNIX/Linux-based and PC-based systems. You can use the CheckSysConf tool to verify that the machine on which

you want to run Cadence tools have the required system configuration and OS patches. Note that this tool is only available for Solaris, HP-UX, AIX, and LINUX platforms. See [Using CheckSysConf](#) for more information.

You can also contact the manufacturer of your platform or Cadence Customer Support at www.cadence.com/support.

- d. Select *Resources - Product Manuals*. The page that appears lists the most frequently used Cadence releases. Click a release to obtain the product manuals pertaining to that release.

You can read the full documentation before installing the products to learn what is new in this release. This library also includes configuration manuals.

If the release you are interested in is not listed, go Search Product Manuals, type "Release Documentation Index Page".

2. Set the default compiler to the standard ANSI C compiler if you have other compilers on your system.

Some Cadence products use the ANSI C compilers to link in customer C code or libraries.

For information on the compilers required for your operating system, contact the manufacturer of your platforms or search in Cadence Online Support.

3. Verify that your system has enough free disk space for the products you are installing.

See your product documentation for more information about memory, disk, and swap requirements for design databases of various sizes.

4. Determine the type of installation.

- If your shipment contains only Baseline or both Baseline, Update, and Hotfix DVDs, follow the instructions in [Chapter 4, "Installing Products into a New Hierarchy."](#)
- If you have previously installed products from this release and now want to install additional products from these DVDs, follow the instructions in [Chapter 5, "Installing Additional Products into an Existing Hierarchy."](#)
- If you have previously installed products from this release and now want to update those products using Update and Hot Fix DVDs, follow the instructions in [Chapter 6, "Updating an Existing Hierarchy."](#)

Note: The type of DVD you mount first (Baseline, Update, and Hotfix DVD#1) determines the resulting installation type. That is, if you start with Baseline DVD#1, you will install the baseline version of the selected products. If you start with an Hotfix DVD#1, you will install the Hotfix version of the products as they existed on the date that the Hotfix DVD set was created. To complete the installation, you may need to mount and install additional products

from the Baseline version after installing from the Hotfix DVDs disk.

Using CheckSysConf

CheckSysConf is a tool that verifies whether the machine on which you want to run Cadence tools has the required system configuration and OS patches. This tool is available for LINUX (32-bit and 64-bit) platforms.

CheckSysConf Components

This tool consists of three parts:

- **A script:** `checkSysConf` is a script that performs the checks.
- **Data files:** A group of data files that define all the parameters and patches required for a particular release of operating system. The script uses these datafiles to determine whether the machine has the required system configuration and system patches.
- **Documentation:** The documentation set consists of the user guide, a README file, and the man pages. The README file and man pages are located in `your_inst_dir/doc/check_conf`.

CheckSysConf Installation Hierarchy

`CheckSysConf` is installed in the following directory structure:

Location	Description
<code>your_inst_dir/tools.lnx86</code>	Contains the LINUX script
<code>your_inst_dir/doc/check_conf</code>	Contains the README file and the man pages
<code>your_inst_dir/share/patchData</code>	Contains the data files

CheckSysConf Syntax

The syntax for running `CheckSysConf` is as follows:

```
checkSysConf release [options]
```

where options can be a combination of one or more of the following:

Option	Description
release	Checks an OS configuration and patch list file located at <code>your_inst_dir/share/patchData/platform type/OS/release</code> for the required system configuration and patches for the Cadence product specified in <code>release</code> .
-c	Checks all existing data files for possible platform/OS conflicts. This option identifies if patches required in one release are known-problem patches in another release.
-h	Prints the help message.
-r	Returns the releases supported in the data directory. The output of this option represents the only valid <code>release</code> option.
-v	Returns the release version of this script.
-i	Suppresses title and other additional information output.
-l	Prints the patch data file associated with <code>platform/OS/release</code> .
-q	Executes the script silently and outputs only <code>PASS</code> or <code>FAIL</code> . This switch will have no effect if used in conjunction with the <code>-l</code> or <code>-p</code> option.
-d	Provides the patch data from the path specified in the <code>directory</code> . The leading portion of the directory can be any valid path, but the last directory in the path must be named <code>patchData</code> .
-p	Prints the names of Cadence products that require this patch.
-noDISPLAY	Allows skipping of the display test, even if there is a <code>DISPLAY</code> requirement in the data file. This argument should be used when running the tool on a remote host, where it is not possible to ascertain the display configuration (or it may not even have one). Using the <code>-noDISPLAY</code> argument prevents a false fail in these circumstances.
--oem	force checkSysConf to use OEM datafile

Note: The output is saved in the `/tmp/checkSysConf.<hostname>-<date>.log` file.

Examples

Example 1

```
checkSysConf <release>
```

Determines if this workstation has the proper patches for <release> name. Eg IC23.1 is a release name.

Example 2

```
checkSysConf <release> -d <patch Directory>
```

Determines if this workstation has the proper patches for <release> based on the patch data files found in the data directory /<installation_dir>/share/patchData. The specified directory must end with the patchData directory, such as IC23.1 -d /home/user/IC231/share/patchData.

Example 3

```
checkSysConf <release> -p <patch>
```

This command will query the data file to determine which Cadence products require a particular patch. This feature allows users or system administrators to ignore a failure if the indicated product is not in use. -

Example 4

```
checkSysConf <release> -q
```

Runs the script silently from another script, which verifies the PASS or FAIL output.

FILES

The data for the platform / OS / patch / release data is contained in a directory. The default location is within the Cadence hierarchy at <installation_dir>/share/patchData

CheckSysConf Data Files

The data files are the actual files against which the configuration and the patch list are checked. By default, the data files except for LINUX are stored in `your_inst_dir/share/patchData`. There is a separate directory for each platform type and release. The actual patch data is in a Cadence tool release file, such as IC23.1, containing the minimum required system configuration and patch levels for that Cadence tool release. For example, the data file for the IC23.1 release on Linux is located at `your_inst_dir/share/patchData/Linux`.

The datafiles for LINUX are stored in

`your_inst_dir/share/patchData/Linux/architecture/vendor/version/data_file`

For example, the datafiles for the IC23.1 release on LINUX on all architecture variants are stored in the following locations

`your_inst_dir/share/patchData/Linux/x86_64/redhat/7.0WS/IC23.1`
`your_inst_dir/share/patchData/Linux/x86_64/redhat/8.0WS/IC23.1`
`your_inst_dir/share/patchData/Linux/ppc64le/redhat/7.0WS/IC23.1`
`your_inst_dir/share/patchData/Linux/ppc64le/redhat/8.0WS/IC23.1`
`your_inst_dir/share/patchData/Linux/x86_64/suse/12/IC23.1`
`your_inst_dir/share/patchData/Linux/x86_64/suse/15/IC23.1`

Data File Format

The data file format for the `checkSysConf` script is different for the LINUX, HP-UX and AIX platforms.

The following apply to all platforms:

- Any line that starts with a `#` is a comment.
- The first line contains the platform name, operating system name, and product release name for which the data file is meant. Note that it is a commented line.
- Each field within a record line must have at least one space separating it from the next field.
- The first field must be one of these keywords: `BANNER`, `MEMORY`, `SWAP`, or `DISPLAY`. There are other keywords, which are specific to platform, such as `HARDWARE`, `LOCALE`, `GRAPHIC-PATCH`, `PATCH`, `NOT-PATCH` (or `FILESET`, `NOT-FILESET` for the IBM platform), and `PACKAGE` for LINUX. The keywords can be divided into two categories:
 - General: which apply to all platform types. The following table describes the general keywords:

Keyword	Description
BANNER	For comments that are to be printed. The script prints out all information between the quotation marks. You should not use quotation marks as part of the comments.

MEMORY & SWAP	For representing value records. The second field on the line must be a number. The implied units are megabytes.
DISPLAY	For checking the display environment of the system. There are four fields for this key: depth (the depth of the visual, such as 8 planes, 24 planes), Class (the class of the visual, such as PseudoColor), Product list (the list of product(s) that requires this check), and for Linux and AIX X-Render requirement (if X-render is required by the product).
EXIT	To be used directly after a BANNER command, so that checksysconf does no further reporting after printing the contents of the banner. This is to prevent important information held in the banner from scrolling off the top of the screen.

- Platform specific. These are described in the following table:

Keyword	Description
PATCH	For required patches to be checked for a particular version excluding the patches for the GRAPHIC card. The keyword is used in Solaris and HP-UX data files.
NOT-PATCH	For patches that need to be removed. The keyword is used in Solaris and HP-UX data files.
PATCH, APAR	Patch indicators and require additional fields representing the APARA numbers that need to be installed. The keyword is used in AIX data files.
FILESET	For required OS filesets for the IBM platform.
NOT- FILESET	For OS filesets that need to be removed.
APAR	For required OS APARs for IBM platform.
PACKAGE	For required LINUX packages, excluding patches for GRAPHIC, HARDWARE, and LOCALE.
NOT- PACKAGE	For packages or package versions that are incompatible with the software. A specific version can be specified.

GRAPHIC-PATCH	For patches specific to a certain graphic card. When the keyword is GRAPHIC-PATCH, the PRODUCT field must contain a known graphic type. These are in the script (more can be added) and currently are: PGX32, CREATOR, ELITE, SX, PGX, CG, and BW.
HARDWARE	For patches specific to a certain hardware model.
LOCALE	For LOCALE patches.
BUNDLE	Present in HP-UX datafiles. It enables product groups to specify a minimum set of software bundles installed on a HP system.

Note: The order of the keyword entries in the file is not important. Remember that there can be only one keyword rule per line.

Sample Data Files

Red Hat 7

For RedHat Linux systems the keywords include a KERNEL entry for determining the specific kernel version in use since this is not always a package but rather a custom-built component.

```
# IC23.1 PM2023 Data File
# RHEL7.4 c05.03 X86_64 data-file
# Cadence IT standard Version 1.4
# Red Hat Enterprise Linux / CentOS Linux 7.4 and all above minor 7.X versions.
```

```
MEMORY 4096
SWAP 4096
XKEYBOARD True
WRITABLE-FILE /dev/shm "edpSharedMemory"
```

```
#KEY    MAJOR MINOR
OSVERSION 7    4
```

```
#KEY    MIN      TITLE      PRODUCT
KERNEL  3.10.0-693  "Kernel Version" (All_Products)
```

```
#-----
# 64-bit system libraries requirements
#-----
```

```
#KEY  PACKAGE      MIN      TITLE      REQUIRED BY      ARCHITECTURE
```

```

PACKAGE glibc-2.17      196.el7  "glibc"          (All_Products)    x86_64
PACKAGE elfutils-libelf-0.168  8.el7  "libelf.so.1"    (All_Products)    x86_64
PACKAGE ksh-20120801    34.el7  "ksh"            (cdnWrapper)      x86_64
PACKAGE redhat-lsb-4.1   27.el7  "lsb_release command" (FlexNet cdnsCompat)
x86_64
PACKAGE xterm-295       3.el7   "xterm command"   (iscap)           x86_64

#KEY PACKAGE MIN TITLE REQUIRED BY ARCHITECTURE
PACKAGE mesa-libGL-17.0.1 6.20170307 "libGL.so.1" (VIRTUOSO) x86_64
PACKAGE mesa-libGLU-9.0.0 4.el7 "libGLU.so.1" (VIRTUOSO) x86_64
PACKAGE motif-2.3.4 12.el7_4 "libXm.so.4" (VIRTUOSO) x86_64
PACKAGE libXp-1.0.2 2.1.el7 "libXp.so.6" (VIRTUOSO) x86_64
PACKAGE libpng12-1.2.50 10.el7 "libpng12.so.0 libpng.so.3" (VIRTUOSO) x86_64
PACKAGE libpng-1.5.13 7.el7_2 "libpng15.so.15" (VIRTUOSO) x86_64
PACKAGE libjpeg-turbo-1.2.90 5.el7 "libjpeg.so.62" (VIRTUOSO) x86_64
PACKAGE expat-2.1.0 10.el7_3 "libexpat.so.1" (VIRTUOSO) x86_64
PACKAGE libXtst-1.2.3 1.el7 "libXtst.so.6" (VIRTUOSO) x86_64
#PACKAGE readline-6.2 10.el7 "libreadline.so.6 libhistory.so.6" (VIRTUOSO) x86_64
#PACKAGE ncurses-libs-5.9 13.20130511.el7 "libncurses.so.5" (VIRTUOSO) x86_64
PACKAGE glibc-devel-2.17 196.el7 "glibc header files" (GCC) x86_64

#-----
# 64-bit system libraries requirements - specific to Virtuoso
#-----

#KEY PACKAGE MIN TITLE REQUIRED BY ARCHITECTURE
# The GDB/pstack package is for crash detector utility (RH 7)
PACKAGE gdb-7.6.1 100 "GNU debugger" (VIRTUOSO) x86_64
PACKAGE libXScrnSaver-1.2.2 6.1.el7 "libXScrnSaver" (VIRTUOSO) x86_64
PACKAGE apr-1.4.8 3 "APR library 64 bits" (VIRTUOSO) x86_64
PACKAGE apr-util-1.5.2 6 "APR Utility library 64 bits" (VIRTUOSO) x86_64
PACKAGE compat-db47-4.7.25 28 "libdb-4.7" (VIRTUOSO) x86_64
PACKAGE xorg-x11-server-Xvfb-1.19.3 11 "Xvfb" (VIRTUOSO) x86_64
# Check for Mesa drivers DRI 3D Viewer
PACKAGE mesa-dri-drivers-17.2.3 8.20171019 "Mesa drivers" (VIRTUOSO) x86_64
PACKAGE openssl-devel-1.0.2k 8 "OpenSSL library" (VIRTUOSO) x86_64
PACKAGE libXext-1.3.3 3 "libXext.so.6" (VIRTUOSO) x86_64
PACKAGE libxkbcommon-x11-0.7.1 1 "libxkbcommon-x11.so.0" (VIRTUOSO)
x86_64
PACKAGE xcb-util-image-0.4.0 2 "libxcb-image.so.0" (VIRTUOSO) x86_64
PACKAGE xcb-util-keysyms-0.4.0 1 "libxcb-keysyms.so.1" (VIRTUOSO) x86_64
PACKAGE xcb-util-renderutil-0.3.9 3 "libxcb-render-util.so.0" (VIRTUOSO) x86_64
PACKAGE xcb-util-wm-0.4.1 5 "libxcb-icccm.so.4" (VIRTUOSO) x86_64

```

```
#-----  
# Display requirement  
#-----  
  
#KEY      DEPTH      CLASS      PRODUCTS  
DISPLAY    8 planes    PseudoColor    (DFII)
```

Red Hat 8

For RedHat Linux systems the keywords include a KERNEL entry for determining the specific kernel version in use since this is not always a package but rather a custom-built component.

```
# IC23.1 PM2023 Data File  
# RHEL8.1 c01.00 X86_64 data-file  
# Cadence IT standard Version 1.1  
# Red Hat Enterprise Linux / CentOS Linux 8.1 and all above minor 7.X versions.  
  
MEMORY 4096  
SWAP 4096  
XKEYBOARD True  
WRITABLE-FILE /dev/shm "edpSharedMemory"  
  
#KEY      MAJOR MINOR  
OSVERSION 8      1  
  
#KEY      MIN      TITLE      PRODUCT  
KERNEL    4.18.0-147  "Kernel Version" (All_Products)  
  
#-----  
# 64-bit system libraries requirements  
#-----  
  
#KEY  PACKAGE      MIN      TITLE      REQUIRED BY      ARCHITECTURE  
PACKAGE glibc-2.28      72.el8_1.1 "glibc"      (All_Products)  x86_64  
PACKAGE elfutils-libelf-0.176 5.el8      "libelf.so.1" (All_Products)  x86_64  
PACKAGE ksh-20120801      253.el8_1 "ksh"      (cdnWrapper)    x86_64  
PACKAGE redhat-lsb-core-4.1  47.el8      "lsb_release command" (FlexNet cdnsCompat)  
x86_64  
PACKAGE xterm-331      1.el8      "xterm command" (iscape)        x86_64  
  
#KEY  PACKAGE      MIN      TITLE      REQUIRED BY      ARCHITECTURE
```

```

PACKAGE mesa-libGL-19.1.4    3.el8_1  "libGL.so.1"      (VIRTUOSO)      x86_64
PACKAGE libglvnd-glx-1.0.1    0.9.git5baa1e5.el8 "libGL.so.1"      (VIRTUOSO)      x86_64
PACKAGE mesa-libGLU-9.0.0     15.el8   "libGLU.so.1"     (VIRTUOSO)      x86_64
PACKAGE motif-2.3.4          16.el8   "libXm.so.4"      (VIRTUOSO)      x86_64
PACKAGE libXp-1.0.3           3.el8    "libXp.so.6"      (VIRTUOSO)      x86_64
PACKAGE libpng12-1.2.57       5.el8    "libpng12.so.0"    (VIRTUOSO)      x86_64
PACKAGE libpng-1.6.34         5.el8    "libpng15.so.15"   (VIRTUOSO)      x86_64
PACKAGE libjpeg-turbo-1.5.3   10.el8   "libjpeg.so.62"    (VIRTUOSO)      x86_64
PACKAGE expat-2.2.5           3.el8    "libexpat.so.1"    (VIRTUOSO)      x86_64
PACKAGE libXtst-1.2.3         7.el8    "libXtst.so.6"     (VIRTUOSO)      x86_64
#PACKAGE readline-7.0         10.el8   "libreadline.so.7 libhistory.so.7" (VIRTUOSO)      x86_64
#PACKAGE ncurses-compat-libs-6.1 7.20180224 "libncurses.so.5" (VIRTUOSO)      x86_64
PACKAGE glibc-devel-2.28      72.el8_1.1 "glibc header files" (GCC)           x86_64

#-----
# 64-bit system libraries requirements - specific to Virtuoso
#-----

#KEY  PACKAGE      MIN  TITLE                REQUIRED BY      ARCHITECTURE
# The GDB/pstack package is for crash detector utility
PACKAGE gdb-8.2        6    "GNU debugger"        (VIRTUOSO)      x86_64
PACKAGE libXScrnSaver-1.2.3 1    "libXScrnSaver"      (VIRTUOSO)      x86_64
PACKAGE apr-1.6.3       9    "APR library 64 bits" (VIRTUOSO)      x86_64
PACKAGE apr-util-1.6.1   6    "APR Utility library 64 bits"(VIRTUOSO)      x86_64
PACKAGE xorg-x11-server-Xvfb-1.20.3 11  "Xvfb"                (VIRTUOSO)      x86_64
# Check for Mesa drivers DFII 3D Viewer
PACKAGE mesa-dri-drivers-19.1.4 3    "Mesa drivers"        (VIRTUOSO)      x86_64
PACKAGE openssl-devel-1.1.1c 2    "OpenSSL library"     (VIRTUOSO)      x86_64
# MISSING compat
PACKAGE libXext-1.3.3     9    "libXext.so.6"        (VIRTUOSO)      x86_64
PACKAGE libxkbcommon-x11-0.8.2 1    "libxkbcommon-x11.so.0" (VIRTUOSO)      x86_64
x86_64
PACKAGE xcb-util-image-0.4.0 9    "libxcb-image.so.0"    (VIRTUOSO)      x86_64
PACKAGE xcb-util-keysyms-0.4.0 7    "libxcb-keysyms.so.1" (VIRTUOSO)      x86_64
PACKAGE xcb-util-renderutil-0.3.9 10   "libxcb-render-util.so.0" (VIRTUOSO)      x86_64
PACKAGE xcb-util-wm-0.4.1    12   "libxcb-icccm.so.4"    (VIRTUOSO)      x86_64

#-----
# Display requirement
#-----

#KEY      DEPTH  CLASS      PRODUCTS
DISPLAY   8 planes  PseudoColor  (DFII)

```

Suse 12

For Suse Linux systems the keywords include a KERNEL entry for determining the specific kernel version in use since this is not always a package but rather a custom-built component.

```
# IC23.1 PM2023 Data File
# SLES12_SP3 c04.00 X86_64 data-file
# Cadence IT standard version 1.0 01/22/2019
# SUSE Enterprise Linux 12 SP3 and all above Service Pack releases.

MEMORY 4096
SWAP 4096
XKEYBOARD True
WRITABLE-FILE /dev/shm "edpSharedMemory"

#KEY      MIN      TITLE      PRODUCT
KERNEL    4.4.73-5    "Kernel Version" (All_Products)

#-----
# 64-bit system libraries requirements
#-----

#KEY  PACKAGE      MIN      TITLE      REQUIRED BY
PACKAGE glibc-2.22    61.3     "glibc"      (All_Products)
PACKAGE libelf1-0.158  6.1      "libelf.so.1" (All_Products)
PACKAGE mksh-50f      5.1      "ksh"        (cdnWrapper)
PACKAGE lsb-release-2.0 28.1     "lsb_release command" (FlexNet cdnsCompat)
PACKAGE xterm-308      3.8      "xterm command" (iscape)

#KEY  PACKAGE      MIN      TITLE      REQUIRED BY
PACKAGE libelf0-0.8.13  18.64    "libelf.so.0" (All_Products)
PACKAGE Mesa-libGL1-17.0.5 116.6    "libGL.so.1" (VIRTUOSO)
PACKAGE libGLU1-9.0.0    18.1     "libGLU.so.1" (VIRTUOSO)
PACKAGE libXm4-2.3.4     4.15     "libXm.so.4" (VIRTUOSO)
PACKAGE libXp6-1.0.2     3.58     "libXp.so.6" (VIRTUOSO)
PACKAGE libpng12-0-1.2.50 19.1     "libpng12.so.0" (VIRTUOSO)
PACKAGE libjpeg62-62.1.0 30.1     "libjpeg.so.62" (VIRTUOSO)
#PACKAGE libreadline6-6.3 82.1     "libreadline.so.6 libhistory.so.6" (VIRTUOSO)
#PACKAGE libncurses5-5.9 40.124   "libncurses.so.5" (VIRTUOSO)
PACKAGE termcap-2.0.8    1042.61  "libcurses.so.1" (VIRTUOSO)
PACKAGE libgdbm4-1.10    9.70     "libgdbm.so.4" (VIRTUOSO)
PACKAGE zlib-devel-1.2.8 11.1     "libz.so"    (VIRTUOSO)
```

```

PACKAGE libexpat1-2.1.0      20.2    "libexpat.so.1"          (VIRTUOSO)
PACKAGE libXtst6-1.2.2      7.1     "libXtst.so.6"          (VIRTUOSO)
PACKAGE glibc-devel-2.22    61.3    "glibc header files"    (GCC)

#-----
# 64-bit system libraries requirements - specific to Virtuoso
#-----

#KEY  PACKAGE      MIN    TITLE                      REQUIRED BY
PACKAGE libapr-util1-1.5.3  1.77    "libaprutil-1.so.0"      (VIRTUOSO)
PACKAGE gdb-8.0            1.9     "gdb"                    (VIRTUOSO)
PACKAGE libopenssl1_0_0-1.0.2j 59.1    "libcrypto.so.1.0.0"    (VIRTUOSO)
PACKAGE libdb-4_8-4.8.30    27.206  "libdb-4.8.so"          (VIRTUOSO)

PACKAGE libXext6-1.3.2      3.61    "libXext.so.6"          (VIRTUOSO)
PACKAGE libxkbcommon-x11-0-0.4.1 1.9     "libxkbcommon-x11.so.0" (VIRTUOSO)
PACKAGE libxcb-image0-0.3.9  8.60    "libxcb-image.so.0"      (VIRTUOSO)
PACKAGE libxcb-keysyms1-0.3.9 8.54    "libxcb-keysyms.so.1"    (VIRTUOSO)
PACKAGE libxcb-render-util0-0.3.8 8.52    "libxcb-render-util.so.0" (VIRTUOSO)
PACKAGE libxcb-icccm4-0.4.0  1.16    "libxcb-icccm.so.4"      (VIRTUOSO)

#-----
# Display requirement
#-----

#KEY      DEPTH    CLASS          PRODUCTS
DISPLAY   8 planes  PseudoColor    (DFII)

```

Suse 15

For Suse Linux systems the keywords include a KERNEL entry for determining the specific kernel version in use since this is not always a package but rather a custom-built component.

```

# IC23.1 PM2023 Data File
# SLES15_SP5 c03.00 X86_64 data-file
# Cadence IT standard Version 1.0
# SUSE Enterprise Linux 15 SP2 and all above Service Pack releases.

MEMORY 4096
SWAP 4096
XKEYBOARD True
WRITABLE-FILE /dev/shm "edpSharedMemory"

#KEY      MIN      TITLE          PRODUCT

```


KERNEL 5.3.18-24.64 "Kernel Version" (All_Products)

#-----

64-bit system libraries requirements

#-----

#KEY	PACKAGE	MIN	TITLE	REQUIRED BY
PACKAGE	glibc-2.26	13.56.1	"glibc"	(All_Products)
PACKAGE	libelf1-0.168	4.5.3	" libelf.so.1 "	(All_Products)
PACKAGE	mksh-56c	1.10	"ksh"	(cdnWrapper)
PACKAGE	lsb-release-3.1	1.16	"lsb_release command"	(FlexNet cdnsCompat)
PACKAGE	xterm-330	9.1	"xterm command"	(iscape)

#KEY	PACKAGE	MIN	TITLE	REQUIRED BY
PACKAGE	Mesa-libGL1-19.3.4	45.23	" libGL.so.1 "	(VIRTUOSO)
PACKAGE	libGLU1-9.0.0	8.45	" libGLU.so.1 "	(VIRTUOSO)
PACKAGE	libXm4-2.3.4	1.33	" libXm.so.4 "	(VIRTUOSO)
PACKAGE	libXp6-1.0.3	1.24	" libXp.so.6 "	(VIRTUOSO)
PACKAGE	libpng12-0-1.2.57	2.18	" libpng12.so.0 "	(VIRTUOSO)
PACKAGE	libjpeg62-62.2.0	5.15.7	" libjpeg.so.62 "	(VIRTUOSO)
#PACKAGE	libreadline7-7.0	9.14.1	" libreadline.so.7 libhistory.so.7 "	(VIRTUOSO)
#PACKAGE	libncurses5-6.1	5.6.2	" libncurses.so.5 "	(VIRTUOSO)
PACKAGE	termcap-2.0.8	1026.33	" libcurses.so.1 "	(VIRTUOSO)
PACKAGE	libgdbm4-1.12	1.418	" libgdbm.so.4 "	(VIRTUOSO)
PACKAGE	zlib-devel-1.2.11	3.21.1	" libz.so "	(VIRTUOSO)
PACKAGE	libexpat1-2.2.5	3.6.1	" libexpat.so.1 "	(VIRTUOSO)
PACKAGE	libXtst6-1.2.3	1.24	" libXtst.so.6 "	(VIRTUOSO)
PACKAGE	libapr-util1-1.6.1	10.21	" libaprutil-1.so.0 "	(VIRTUOSO)
PACKAGE	glibc-devel-2.26	13.56.1	"glibc header files"	(GCC)

#-----

64-bit system libraries requirements - specific to Virtuoso

#-----

#KEY	PACKAGE	MIN	TITLE	REQUIRED BY
PACKAGE	gdb-8.3.1	0	"GNU debugger"	(VIRTUOSO)
PACKAGE	xorg-x11-server-1.20.3	0	"Xorg X server"	(VIRTUOSO)
PACKAGE	libopenssl1_0_0-1.0.2p 0	0	" libcrypto.so.1.0.0 libssl.so.1.0.0 "	(VIRTUOSO)
# compat				
PACKAGE	libdb-4_8-4.8.30	0	" libdb-4.8.so "	(VIRTUOSO)
PACKAGE	libXext6-1.3.3	1.30	" libXext.so.6 "	(VIRTUOSO)
PACKAGE	libxkbcommon-x11-0-0.8.2	3.3.1	" libxkbcommon-x11.so.0 "	(VIRTUOSO)
PACKAGE	libxcb-image0-0.4.0	1.23	" libxcb-image.so.0 "	(VIRTUOSO)
PACKAGE	libxcb-keysyms1-0.4.0	1.23	" libxcb-keysyms.so.1 "	(VIRTUOSO)
PACKAGE	libxcb-render-util0-0.3.9	1.23	" libxcb-render-util.so.0 "	(VIRTUOSO)

PACKAGE libxcb-icccm4-0.4.1 1.24 "[libxcb-icccm.so.4](#)" (VIRTUOSO)

#-----

Display requirement

#-----

#KEY	DEPTH	CLASS	PRODUCTS
DISPLAY	8 planes	PseudoColor	(DFII)

Installing Products into a New Hierarchy

To install, configure, and test your Cadence® products and licensing for a new release,

1. Do one of the following:
 - If you received an `Installation Control` file for this release via e-mail, save it to a known location.
2. Verify that you have read permission for the `Installation Control` file.
3. Save the ASCII license file that you received as an email attachment in a known location and ensure that you have read permission.
4. Identify the DVD to use for installation.

If you have received only a set of Baseline DVDs, use the Baseline DVD#1 first. The disk should be labeled `Disk # of n for platform`.

If you have received both a set of Base, Update, and Hotfix DVDs (and the `Installation Control` e-mail file), use the Update and Hotfix DVD#1 first. After installing the Update or Hotfix DVDs, InstallScape may ask you to then mount the baseline DVD to complete the installation in a new hierarchy.

Note: For Windows users, the Update/Hotfix DVDs contains the full release. The baseline DVD will not be required to complete the installation in a new hierarchy.

Note: The type of DVD you mount first (Baseline, Update, and Hotfix DVD#1) determines the resulting installation type. That is, if you start with Baseline DVD#1, you will install the baseline version of the selected products. If you start with an Hotfix DVD#1, you will install the Hotfix version of the products as they existed on the date that the Hot Fix DVD set was created.

5. Mount the DVD.

Note: The installation starts automatically when you insert the DVD on a Microsoft Windows system. If you are using a dongle, attach it to your computer before inserting the DVD.

- a. Log in as `root` or use `sudo`.

Note: In instance if you DVD drive does not automount, you will need root permission to

mount a DVD.

- b. If it does not already exist, create the DVD directory by typing

```
mkdir /dvd_dir
```

- c. Insert the DVD into the DVD drive.
- d. Mount the DVD drive on the DVD directory, using the appropriate command:

Linux

```
mount -o ro /dev/device /dvd_dir
```

Note: Typically, the device name to represent DVD in Linux is dvd.

- e. Log out as `root`.

6. Check the top level of your DVD for `README` file or *READMEs* directory.

If this file exists, it contains information that might be necessary for the installation of your product.

7. Start the interactive setup utility, `SETUP.SH`.

On a UNIX system, type

```
dvd_mount_point/SETUP.SH
```

For example, if `dvd_dir` is the DVD mount point, then the command to run `SETUP.SH` will be

```
/dvd_dir/SETUP.SH
```

On a Microsoft Windows system, the installation should start automatically. If it does not, go to the top-level directory of the DVD and run the `setup.exe` program.

8. When `SETUP.SH` asks you to specify the path to *your_install_dir*, do one of the following:
- Type the path to an empty existing directory.
 - Type a name for a directory (`SETUP.SH` will create the path for you).
9. Answer the prompts, and proceed as directed.

The appropriate installation utility, `InstallScape` or `InstallShield`, should start automatically.

On Windows, double-click on `setup.exe` at the top level of the DVD.

10. Do one of the following:

- Load the products and licensing using `InstallShield`.
 - `InstallShield` is a GUI-based installation wizard.

- Each screen includes instructions. InstallShield allows you to install the license manager, install products, and update your license file.
 - The installation is customized for each product family: see the `README` file or *README*S directory at the top level of your DVD for appropriate information.
- Load the products using InstallScape. See the InstallScape User Guide for details on how to use InstallScape for installing products.

You will need to configure the license file before you can start using it. For more information, see the Cadence License Manager. This document is available on Cadence Online Support or in the Installation with the product manuals. Or, if you have a previously licensed Cadence hierarchy available, you can read this document using Cadence Help. If your installation consists of more than one DVD, either because the release is on two volumes or because you began the installation with a Update and Hotfix DVDs DVD, InstallScape installs the selected products from the first disk and then prompts you to dismount that DVD and continue the installation using the next disk. For Update and Hotfix installations, you finish by installing the non-updated products from the baseline DVD.

Creating the Tools Link (if required)

Some Cadence tools require a tools directory to be present in the installation hierarchy. This tools directory should be a symbolic link to the `tools.<platform>` directory that is created at the time of installation. Typically, this link is created automatically when you run product configuration scripts such as licensing configuration.

If your product requires this link and if this link does not exist, do the following:

1. Change to the installation directory by specifying the following command at the command prompt:

```
cd <install_dir>
```

2. Create the tools link by specifying the following command at the command prompt:

```
ln -s tools.<platform> tools
```

Installing Additional Products into an Existing Hierarchy

You can install software from only a single release in a given installation directory. InstallScape does not let you install products from different releases in the same hierarchy.

You can add products using Update and Hotfix DVDs or Baseline DVDs. You can add products using a DVD set that was the same release identified as the release in the hierarchy.

Note: If you are adding products using Update and Hotfix DVDs, you will also need the corresponding Baseline DVD set.

Protecting Your Production Environment

To protect your production environment during the installation process,

- Do not install your software in a directory that is your current production hierarchy. Install your software in a new hierarchy or a copy of your production hierarchy.
- Do not install software from more than one baseline release into the same hierarchy.
- Test the installation fully before making it your production hierarchy.

To add products into an existing hierarchy,

1. Log in to the proper user account.
2. Create a duplicate hierarchy from the original Baseline DVD using the same method you used to install the production hierarchy.

Alternatively, you can create a duplicate of the Cadence hierarchy you want to update by using operating system commands such as the UNIX/Linux `tar` command.

Do not use the `cp` command because it does not preserve file system links.

3. Identify the DVD to use first.

If you have received only a set of Baseline DVDs, use the Baseline DVD#1 first.

If you have received only a set of Update and Hotfix DVDs (and the `Installation Control e-mail` file), use the Update and Hotfix DVD#1 first.

Note: You will need the corresponding Baseline DVD set to complete installation. The InstallScape utility asks you to mount the baseline DVD after installing the Update and Hot Fix DVD.

If you have received both a set of Baseline and Update or Hotfix DVDs (and the `Installation Control e-mail` file), use the Update or Hotfix DVD#1 first.

Note: The type of DVD you mount first (Baseline, Update or Hotfix DVD#1) determines the resulting installation type. That is, if you start with Baseline DVD#1, you will install the Baseline version of the selected products. If you start with an Hotfix DVD#1, you will install the Hotfix version of the products as they existed on the date that the Hotfix DVD set was created.

4. Mount the DVD.

Note: The installation starts automatically when you insert the DVD on a Microsoft Windows system.

- a. Log in as `root` or use `sudo`.

Note: You do not normally need root permission to mount a DVD but in case the system does not automount you can manual mount the DVD.

- b. If it does not already exist, create the DVD directory by typing

```
mkdir /dvd_dir
```

- c. Insert the DVD into the DVD drive.

- d. Mount the DVD drive on the DVD directory, using the appropriate command:

Windows	Windows automatically mount the DVD
Linux	<code>mount -o ro /dev/dvd /dvd_dir</code>

- e. Log out as `root`.

5. Start the interactive setup utility, `SETUP.SH`.

On a UNIX system, type

```
dvd_mount_point/SETUP.SH
```

On a Microsoft Windows system, the installation should start automatically. If it does not, go to the top-level directory of the DVD and run the `setup.exe` program.

6. When `SETUP.SH` asks you to specify the path to `your_install_dir`, type the path to an existing Cadence installation directory (this could be the duplicate hierarchy you created previously).
7. Answer the prompts, and proceed as directed.

The appropriate installation utilities, InstallScape or InstallShield, should start automatically. If it does not start, do one of the following:

- On UNIX, type

```
./iscape/bin/iscape.sh&
```

- On Windows, double-click on `setup.exe` at the top level of the DVD.

8. Do one of the following:

- Load the Products and Licensing Using InstallShield.

InstallShield is a GUI-based installation wizard. Each screen includes instructions.

InstallShield allows you to install the license manager, install products, and update your license file.

The installation is customized for each product family: see the `README` file at the top level of your DVD for appropriate information.

- Load the Products Using InstallScape.

If InstallScape has not started, type

```
./iscape/bin/iscape.sh&
```

See the InstallScape User Guide for more details on how to use InstallScape for installing Cadence products.

Creating the Tools Link (if required)

Some Cadence tools require a tools directory to be present in the installation hierarchy. This tools directory should be a symbolic link to the `tools.<platform>` directory that is created at the time of installation. Typically, this link is created automatically when you run product configuration scripts such as licensing configuration.

If this link does not exist, do the following:

1. Change to the installation directory by specifying the following command at the command

prompt:

```
cd <install_dir>
```

2. Create the tools link by specifying the following command at the command prompt:

```
ln -s tools.<platform> tools
```

Updating an Existing Hierarchy

In a given installation directory, you can install software only from a single release. InstallScape does not let you install products from different releases in the same hierarchy.

You can update a hierarchy using a Baseline DVD or an Update or Hotfix DVD. You can update a hierarchy with a DVD set that has the same release identifier as the release identifier of the release installed in the hierarchy.

Protect Your Production Environment

To protect your production environment during the installation process,

- Do not install your software in a directory that is your current production hierarchy. Install your software in a new hierarchy or a copy of your production hierarchy.
- Do not install software from more than one baseline release into the same hierarchy.
- Test the installation fully before making it your production hierarchy.

Updating the Existing Hierarchy

To install the release into an existing hierarchy,

1. Log in to the proper user account.
2. Create a duplicate hierarchy from the original Baseline DVD using the same method you used to install the production hierarchy.

Alternatively, you can create a duplicate of the Cadence hierarchy you want to update by using operating system commands, such as the UNIX/Linux `tar` command.

Do not use the `cp` command because it does not preserve file system links.

3. Identify the DVD to use first.

If you have received only a set of Baseline DVDs, use the Baseline DVD#1 first.

If you have received only a set of Update or Hotfix DVDs (and the `Installation Control e-`

mail file), use the Update or Hotfix DVD#1 first.

Note: You will need the corresponding Baseline DVD set to complete the installation.

If you have received both a set of Baseline and Update or Hotfix DVDs (and the `Installation Control` e-mail file), use the Update or Hotfix DVD#1 first.

Note: The type of DVD you mount first (Baseline, Update, or Hotfix DVD#1) determines the resulting installation type. That is, if you start with Baseline DVD#1, you will install the baseline version of the selected products. If you start with an Hotfix DVD#1, you will install the Hot Fix version of the products as they existed on the date that the Hotfix DVD set was created.

4. Mount the DVD.

Note: The installation starts automatically when you insert the DVD on a Microsoft Windows system.

- a. Log in as `root` or use `sudo`.

Note: You do not normally need root permission to mount a DVD but in case your system does not automount you have to manually mount the DVD.

- b. If it does not already exist, create the DVD directory by typing

```
mkdir /dvd_dir
```

- c. Insert the DVD into the DVD drive.

- d. Mount the DVD drive on the DVD directory, using the appropriate command:

Windows	Windows automatically mount the DVD
Linux	<pre>/bin/mount -o ro /dev/<device> /dvd_dir</pre> Note: Typically, the device name to represent DVD in Linux is <code>dvd</code>.

- e. Log out as `root`.

5. Start the interactive setup utility.

On a UNIX or Linux system, type

```
dvd_mount_point/SETUP.SH
```

On a Microsoft Windows system, the installation should start automatically. If it does not, go to the top-level directory of the DVD and run the `setup.exe` program.

6. When SETUP.SH asks you to specify the path to *your_install_dir*, type the path to an existing Cadence installation directory (this could be the duplicate hierarchy you created previously).
7. Answer the prompts, and proceed as directed.

The appropriate installation utilities, InstallScape or InstallShield, should start automatically. If it does not start, do one of the following:

- On UNIX/Linux, type

```
./iscape/bin/iscape.sh&
```

- On Windows, double-click on `setup.exe` at the top level of the DVD.

8. Do one of the following:

- Load the Products and Licensing Using InstallShield.

InstallShield is a GUI-based installation wizard. Each screen includes instructions.

InstallShield allows you to install the license manager, install products, and update your license file.

The installation is customized for each product family: see the `README` file at the top level of your DVD for appropriate information.

- Load the Products Using InstallScape.

If InstallScape has not started, type

```
./iscape/bin/iscape.sh&
```

See the InstallScape User Guide for more details on how to use InstallScape for installing Cadence products.

Creating the Tools Link (if required)

Some Cadence tools require a tools directory to be present in the installation hierarchy. This tools directory should be a symbolic link to the `tools.<platform>` directory that is created at the time of installation. Typically, this link is created automatically when you run product configuration scripts such as licensing configuration.

If this link does not exist, do the following:

1. Change to the installation directory by specifying the following command at the command prompt:

```
cd <install_dir>
```

2. Create the tools link by specifying the following command at the command prompt:

```
ln -s tools.<platform> tools
```

Advanced Installation Topics

Automating an InstallScape Installation

See the InstallScape BatchMode User Guide for details on how to automate product installation using InstallScape command line options.

Install from Cadence Server without Saving an Archive option, enter the following at the terminal prompt:

- `iscape.sh -batch majorAction=download minorAction=Install
sourceLocation=https://sw.cadence.com/is/IC231/lnx86/Base
installDirectory=/vols/Cadence/IC231`

Enter your username and password as prompted to begin the installation.

Note the URL for sourceLocation is in the Control file email. Simple prepend <https://sw.cadence.com/> to the SOURCE `/is/IC231/lnx86/Base` to form the complete software URL path. The installDirectory is your install directory path. The install directory must be empty and that you have permission to write into.

To get a complete InstallScape software sourceLocation, enter the following at the terminal prompt:

- `iscape.sh -batch majorAction=selectReleases minorAction=getSourceSites
platforms=lnx86 releases=IC231`

This command options return of the URLs of the IC231 release name.

Printing out list of available source sites:

```
https://sw.cadence.com/is/IC231/lnx86/Base  
https://sw.cadence.com/is/IC231/lnx86/Hotfix_IC23.10.030
```

For more command line options go to <https://downloads.cadence.com> select DOWNLOAD ASSISTANTS ->InstallScape and open the *Batch_Mode.pdf*

Configure the software after installing, enter the following at the terminal prompt:

- `iscape.sh -batch majorAction=configure installDirectory=/vols/Cadence/IC231`

If the software release has a configuration built in, the configuration script will be added and display on terminal to run. For example, the batch configure above added an entry below

Adding to configuration script: SftShrlics

```
Initializing install directory to: /vols/Cadence/IC231
Selected release is IC231_lnx86
```

```
CREATING configuration script:
/vols/Cadence/IC231/installData/IC231_lnx86/batch_configure.sh
```

The following products will be added to the configuration script:

```
Adding to configuration script: ncvhdl64b
Adding to configuration script: ncvhdl
Adding to configuration script: amsEnv64b
Adding to configuration script: ihdl64b
Adding to configuration script: oaRedis
```

Done adding products to configuration script.

```
IMPORTANT: To configure products run the script:
/bin/sh /vols/Cadence/IC231/installData/IC231_lnx86/batch_configure.sh
```

For more command line options go to <https://downloads.cadence.com> select DOWNLOAD ASSISTANTS ->InstallScape and open the Batch_Mode.pdf

Get the latest License Manager, InstallScape, and CheckSysConf

Go to <https://downloads.cadence.com>, enter *LCU* in the Filter by Release and select **Lic+Config_Utils (LCU)** to download. Unpack and install the release

Alternative if you have InstallScape run `iscape.sh`, click on **"Search & Install Releases"** to connect to Cadence server. Under the Filter releases by release name scroll down to LCU43, select it, choose the port, and select the latest Hotfix. Click the checkbox the most current Release Date and click Continue and follow the wizard to install the release

Overview of Software Configuration

Before you can run or configure Cadence® products, you must install and configure licensing.

This section covers the following topics:

- [About License Servers](#)
- [About Your Work Order and License File](#)
- [Installing and Configuring Licensing with InstallScope](#)
- [Default and Customized Licensing](#)
- [Licensing Tips](#)

About License Servers

A license server or standalone workstation must

- Be a supported platform and operating system (it cannot be a clone)
- Have a host ID that matches the host ID in the license file
- Have a local disk that contains the UNIX operating system and UNIX file systems, such as `/usr/tmp` (a license server cannot be diskless)
- Have TCP/IP running
- Have license daemons running

About Your Work Order and License File

Your work order includes a license file for your license server's host ID. Depending on the hardware platform, the host ID is the PROM ID, Ethernet address, or another unique identifier for the workstation.

Host IDs are case-insensitive, but host IDs in the license file must match the host IDs of the license servers. In fault-tolerant licensing, the three host IDs in the license file must match the host IDs of

the three license servers.

Installing and Configuring Licensing with InstallScape

The *Cadence License Manager Guide* fully describes the licensing environment and includes instructions on configuring licensing. The guide is available in Cadence Help and in Cadence Online Support (under "Installation," click *Installation, Licensing & System Configuration information*).

To configure licensing from Cadence InstallScape,

1. If you have already been using Cadence products, copy to your new `your_install_dir/share/license` directory any files you customized for your older software, such as a clients file or a license file containing valid temporary licenses.

Your current temporary licenses are merged into your new license file if

- Your temporary licenses are for the same license server
- Your temporary licenses are still valid
- You configure licensing from InstallScape or with `configure` or `lic_config`

As the utilities proceed, you see several different types of license files depending on any previous licensing.

2. From the InstallScape Main Menu, click *Configure*, then click *System Environment*.
3. Choose the *SoftShare Licensing Server* (`host_id`) to configure.

Respond to the prompts.

If you are setting up fault-tolerant licensing, do not start the license daemons when prompted.

4. Back up the `your_install_dir/share/license` files you customized.
5. If you are setting up fault-tolerant licensing, follow the steps in the *Cadence License Manager Guide*.
6. If the license file is in ASCII,
 - a. Copy the license file to `your_install_dir/share/license`.
 - b. Configure the license file (`your_install_dir/share/license/configure`)

Default and Customized Licensing

If you are installing Cadence products for the first time in this hierarchy, use InstallScape to configure Cadence SoftShare licensing. InstallScape configures SoftShare licensing and work orders by

- Creating a link between the `your_install_dir/tools.platform` and `your_install_dir/tools` directories
- Editing the `SERVER` and `DAEMON` lines in the license file
- Creating a script to start the license daemons
- Editing your workstation's boot script to run the license daemon start-up script
- Editing a Cadence-specific file that is used to locate the license files client workstations use

InstallScape runs the licensing configuration script, which lets you choose a default licensing configuration or override any of the defaults. The *Cadence License Manager Guide* describes your options. You can use the default licensing configuration when

- The user workstation is the license server
- The licensing log file is `/usr/tmp/license.log`
- The license server does not use an options file (described in the *Cadence License Manager Guide*)
- Applications use a `clients` file to locate their license server

The default `clients` file lets all workstations use the software controlled by the license file.

- You want to start the `lmgrd` daemon without options using the following command:

```
your_install_dir/tools/bin/lmgrd -c license_file > /usr/tmp/license.log
```

You can minimize the chance of users inadvertently shutting down the license daemons by starting the `lmgrd` daemon with one of these methods instead of using the default options:

- `lmgrd -2 -p`

When you use this command, only members of the `lmadmin` group can run `lmdown`, `lmremove`, and `lmreread`. If `root` should be able to use `lmdown`, `root` must be in the `lmadmin` group. If no `lmadmin` group exists, only a user belonging to the `wheel` group can use these utilities. (Typically, `root` is in `wheel`.)

- `lmgrd -x lmdown`

When you use this command, no one, not even `root`, can run `lmdown`. The license daemons can only be stopped with `kill`. Do not use `kill -9`.

Note: Do not kill the license daemons while licenses are in use because of possible loss of data.

- `lmgrd -x lmremove`

When you use this command, no one, not even `root`, can run `lmremove` to return licenses to the license pool.

Licensing Tips

Here are some tips on setting up and configuring licensing:

- InstallScape can configure licensing only from work orders or e-mail files.
- You can choose the license file in `your_install_dir/share/license` that you want to configure.
- You must configure SoftShare licensing before you configure your products and libraries and before you use Cadence software.
- If you need more information about a particular screen, click *Help* or see the *Cadence License Manager Guide*.
- If you are setting up fault-tolerant licensing, do not start the license daemons when prompted.
- To start the license daemons, you must have write permission for the licensing debug log file.
- Some Cadence products use the `clients` file to locate the correct license file. If you did not override the `clients` file defaults when configuring licensing, all workstations can access the `license` file. The online *Cadence License Manager* guide describes other methods of locating license files and modifying the `clients` file.

Overview of User Environment Configuration

All users must include the path to the Cadence software in their search paths.

To set up user environments,

- For the C shell
 - In the user's `.cshrc` file, set the search path as follows:

```
set path = ($path your_install_dir/tools/bin)
```

Note: Do not set the `DISPLAY` variable in the `.cshrc` file. Doing so can cause licensing and other problems when running remote applications.

- Source the user's `.cshrc` file.
- For the Bourne or Korn shell
 - Add the following lines to the `.profile` file:

```
PATH=$PATH:/your_install_dir/tools/bin
export PATH
```

- Force the new lines to be read by typing

```
. ./profile
```

Overview of Client Workstation Configuration

If the Cadence software does not reside on user workstations, you must set up the user workstations because client workstations must access the software on the file server.

- Check for adequate memory and swap space.

To enhance performance, verify that each client workstation has enough memory (RAM) and swap space. For memory and swap space requirements, see your product documentation.

- Make sure users can access the license server from the their workstations.
- Make sure the user can access the Cadence software.
- Verify the correct operating system and patches are on the client workstation.

Troubleshooting

Go to [Cadence Online Support Portal](#) for troubleshooting tips related to *Installation and Licensing*.